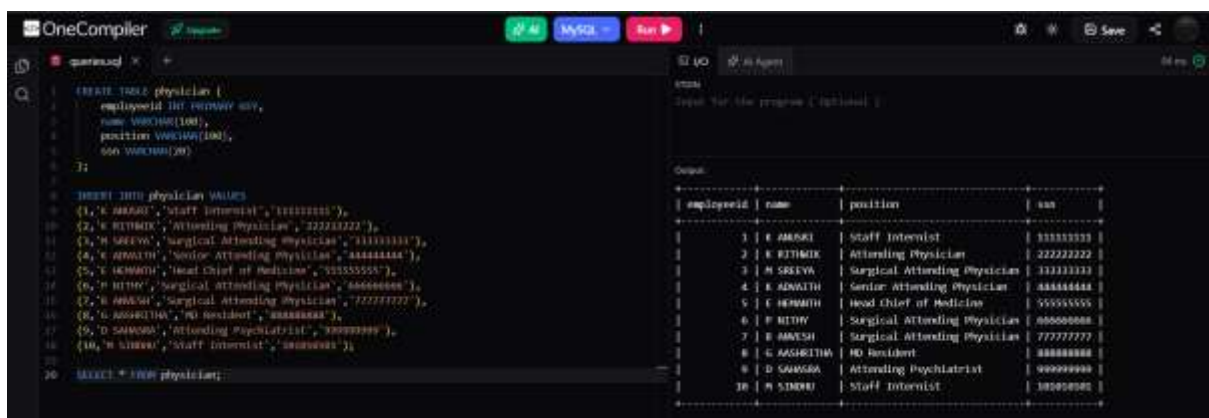


SQL Practice — Physician & Department Tables

SETUP CODES

STEP 1 + STEP 2 — Create Physician Table and Insert Data



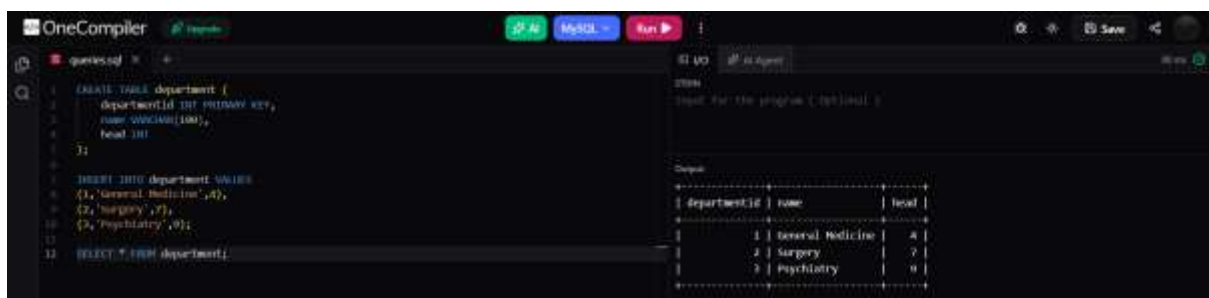
```
CREATE TABLE physician (
  employeeid INT PRIMARY KEY,
  name VARCHAR(100),
  position VARCHAR(100),
  ssn VARCHAR(20)
);

INSERT INTO physician VALUES
(1, 'K. ARAGAO', 'Staff Internist', '111111111'),
(2, 'K. RITHMIR', 'Attending Physician', '222222222'),
(3, 'M. SREYNA', 'Surgical Attending Physician', '333333333'),
(4, 'K. ADNATH', 'Senior Attending Physician', '444444444'),
(5, 'K. ADNATH', 'Head Chief of Medicine', '555555555'),
(6, 'P. MITW', 'Surgical Attending Physician', '666666666'),
(7, 'E. ANVESH', 'Surgical Attending Physician', '777777777'),
(8, 'G. ANSHRITHA', 'MD Resident', '888888888'),
(9, 'D. SAMOSA', 'Attending Psychiatrist', '999999999'),
(10, 'M. SINDHU', 'Staff Internist', '101010101');

SELECT * FROM physician;
```

employeeid	name	position	ssn
1	K. ARAGAO	Staff Internist	111111111
2	K. RITHMIR	Attending Physician	222222222
3	M. SREYNA	Surgical Attending Physician	333333333
4	K. ADNATH	Senior Attending Physician	444444444
5	K. ADNATH	Head Chief of Medicine	555555555
6	P. MITW	Surgical Attending Physician	666666666
7	E. ANVESH	Surgical Attending Physician	777777777
8	G. ANSHRITHA	MD Resident	888888888
9	D. SAMOSA	Attending Psychiatrist	999999999
10	M. SINDHU	Staff Internist	101010101

STEP 3 + STEP 4 — Create and Insert Department Data



```
CREATE TABLE department (
  departmentid INT PRIMARY KEY,
  name VARCHAR(100),
  head INT
);

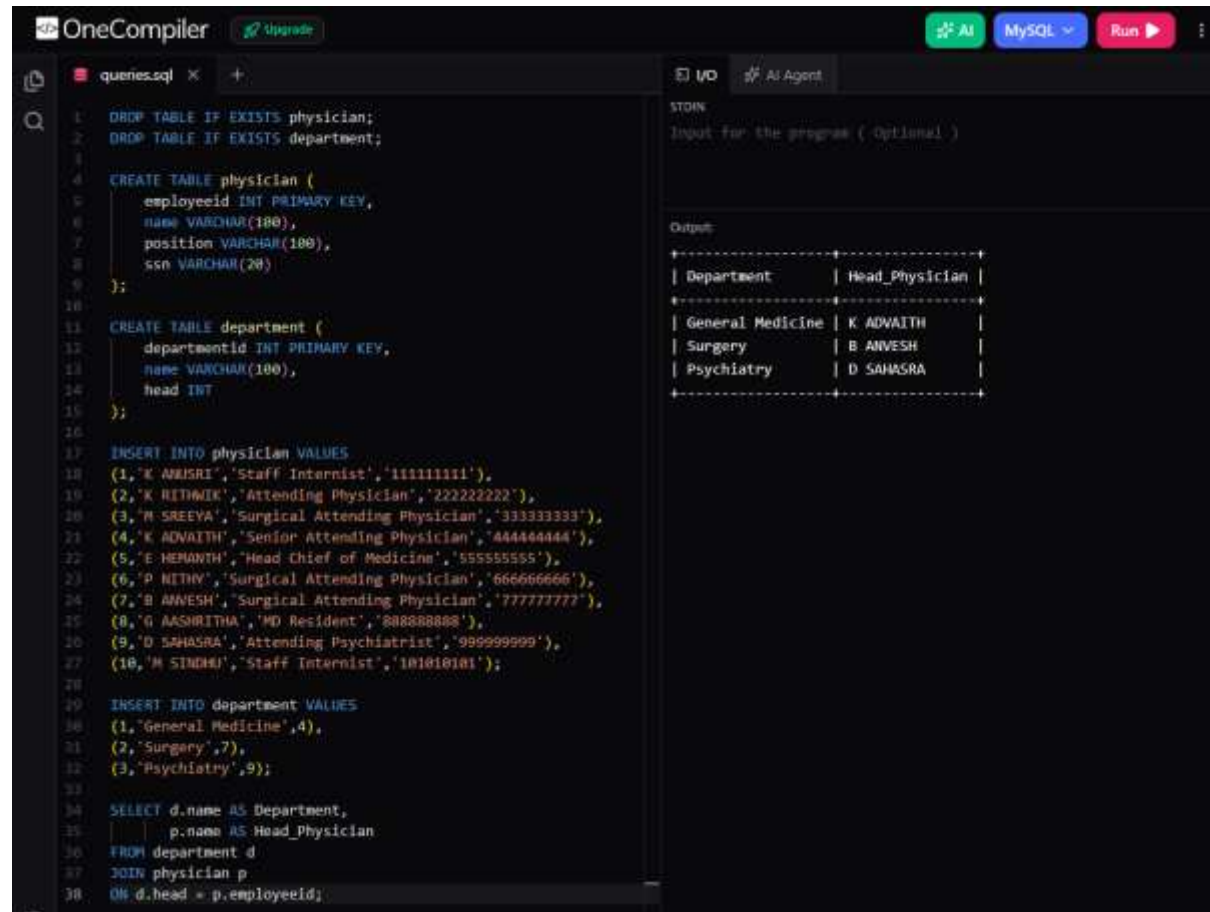
INSERT INTO department VALUES
(1, 'General Medicine', 4),
(2, 'Surgery', 7),
(3, 'Psychiatry', 9);

SELECT * FROM department;
```

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

Q1 — Display Each Department and Its Department Head

INNER JOIN — Department and Department Head



The screenshot shows the OneCompiler IDE with a MySQL database. The code in the editor defines two tables, `physician` and `department`, and performs an inner join to display department heads.

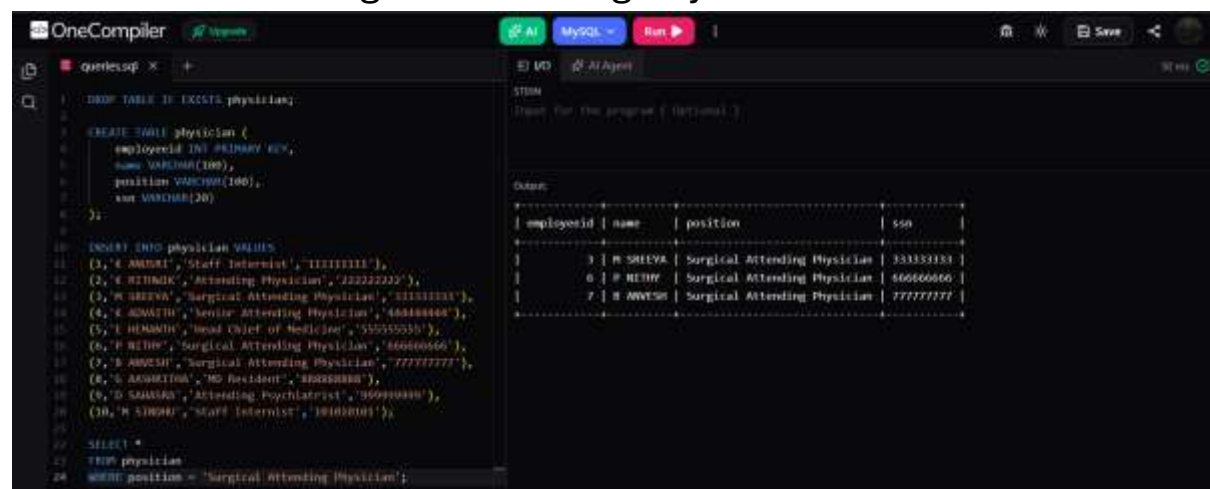
```
1 DROP TABLE IF EXISTS physician;
2 DROP TABLE IF EXISTS department;
3
4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10
11 CREATE TABLE department (
12     departmentid INT PRIMARY KEY,
13     name VARCHAR(100),
14     head INT
15 );
16
17 INSERT INTO physician VALUES
18 (1,'K AMISRI','Staff Internist','111111111'),
19 (2,'K RITHVIK','Attending Physician','222222222'),
20 (3,'M SREEYA','Surgical Attending Physician','333333333'),
21 (4,'K ADVAITH','Senior Attending Physician','444444444'),
22 (5,'E HERMANTH','Head Chief of Medicine','555555555'),
23 (6,'P NITHY','Surgical Attending Physician','666666666'),
24 (7,'B ANVESH','Surgical Attending Physician','777777777'),
25 (8,'G AASHRITHA','MD Resident','888888888'),
26 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
27 (10,'M SINDHU','Staff Internist','101010101');
28
29 INSERT INTO department VALUES
30 (1,'General Medicine',4),
31 (2,'Surgery',7),
32 (3,'Psychiatry',9);
33
34 SELECT d.name AS Department,
35        p.name AS Head_Physician
36 FROM department d
37 JOIN physician p
38 ON d.head = p.employeeid;
```

The output shows the result of the query:

Department	Head_Physician
General Medicine	K ADVAITH
Surgery	B ANVESH
Psychiatry	D SAHASRA

Q2 — WHERE Clause

WHERE — Find Surgical Attending Physicians



The screenshot shows the OneCompiler IDE with a MySQL database. The code in the editor defines the `physician` table and filters for surgical attending physicians.

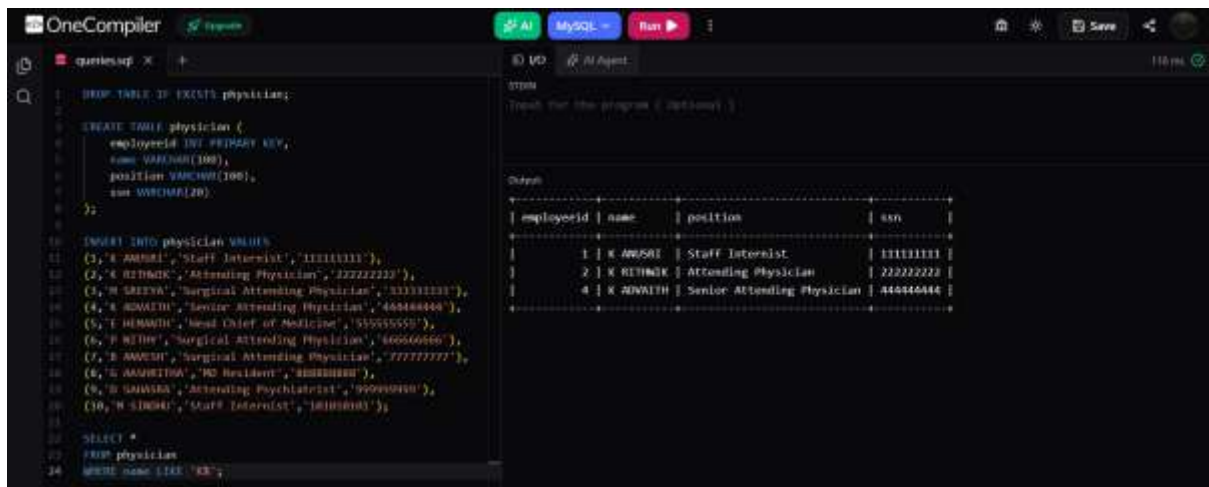
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K AMISRI','Staff Internist','111111111'),
12 (2,'K RITHVIK','Attending Physician','222222222'),
13 (3,'M SREEYA','Surgical Attending Physician','333333333'),
14 (4,'K ADVAITH','Senior Attending Physician','444444444'),
15 (5,'E HERMANTH','Head Chief of Medicine','555555555'),
16 (6,'P NITHY','Surgical Attending Physician','666666666'),
17 (7,'B ANVESH','Surgical Attending Physician','777777777'),
18 (8,'G AASHRITHA','MD Resident','888888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
20 (10,'M SINDHU','Staff Internist','101010101');
21
22 SELECT *
23 FROM physician
24 WHERE position = 'Surgical Attending Physician';
```

The output shows the result of the query:

employeeid	name	position	ssn
3	M SREEYA	Surgical Attending Physician	333333333
6	P NITHY	Surgical Attending Physician	666666666
7	B ANVESH	Surgical Attending Physician	777777777

Q3 — WHERE + LIKE

LIKE — Find Physicians Whose Names Start with K



The screenshot shows the OneCompiler MySQL interface. The left pane contains a SQL script with the following code:

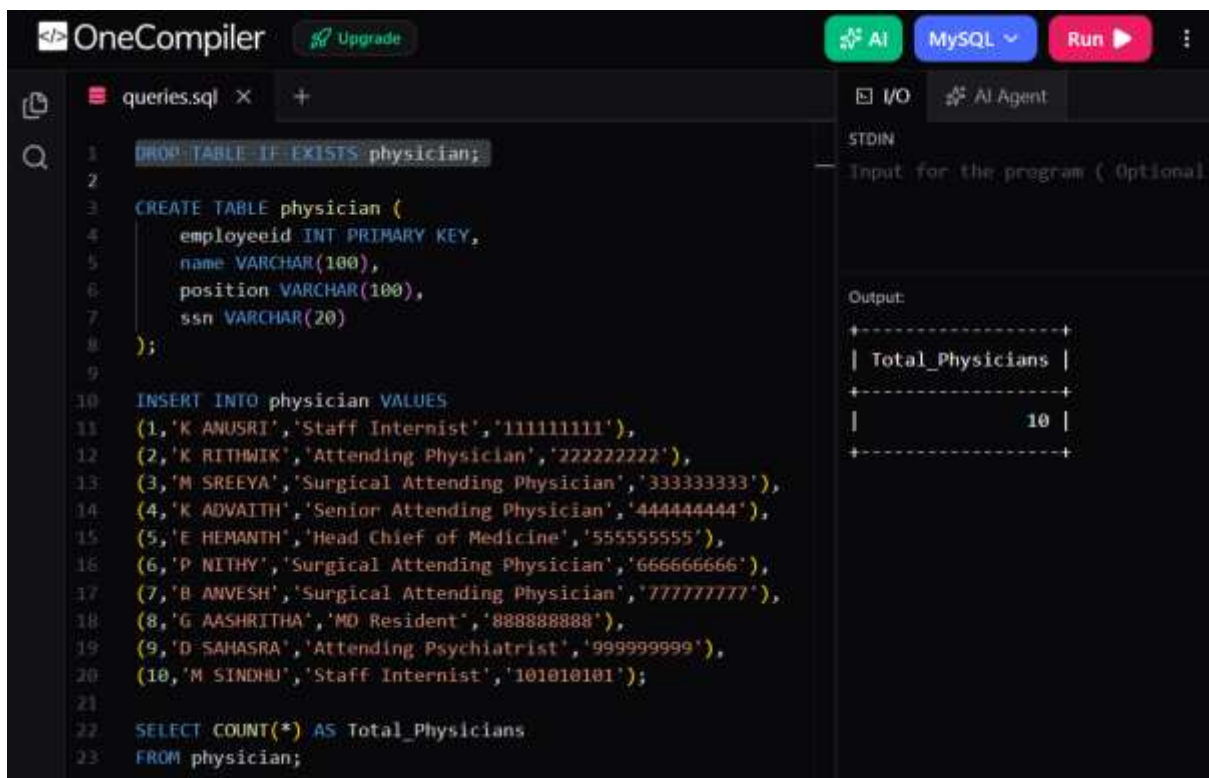
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHWIK','Attending Physician','22222222'),
13 (3,'M SREEYA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SINDHU','Staff Internist','10101010');
21
22 SELECT *
23 FROM physician
24 WHERE name LIKE 'K%';
```

The right pane shows the output of the query:

employeeid	name	position	ssn
1	K ANUSRI	Staff Internist	11111111
2	K RITHWIK	Attending Physician	22222222
4	K ADVAITH	Senior Attending Physician	44444444

Q4 — COUNT()

COUNT — Count the Total Number of Physicians



The screenshot shows the OneCompiler MySQL interface. The left pane contains a SQL script with the following code:

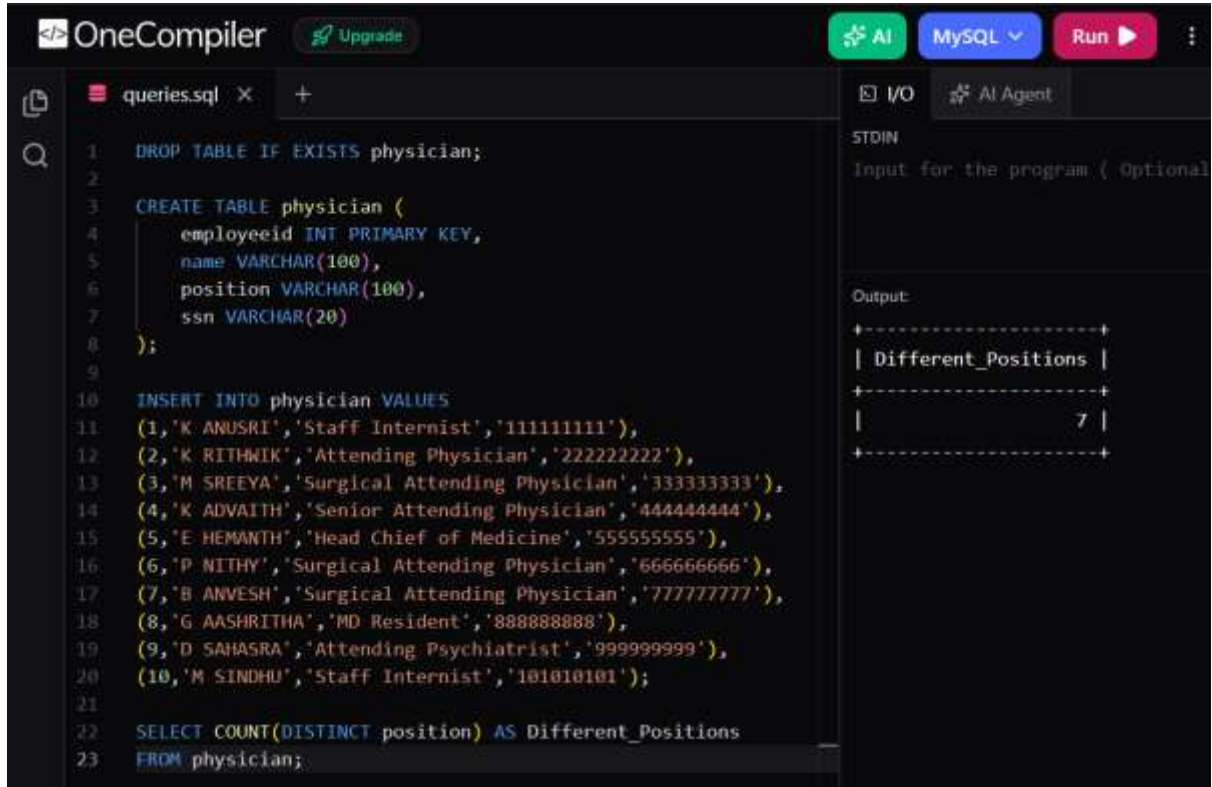
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHWIK','Attending Physician','22222222'),
13 (3,'M SREEYA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SINDHU','Staff Internist','10101010');
21
22 SELECT COUNT(*) AS Total_Physicians
23 FROM physician;
```

The right pane shows the output of the query:

Total_Physicians
10

Q5 — COUNT DISTINCT

COUNT DISTINCT — Count Different Physician Positions



The screenshot shows the OneCompiler MySQL interface. The main editor contains a SQL script that creates a table named 'physician' with columns 'employeeid', 'name', 'position', and 'ssn'. It then inserts 10 rows of data with various physician names and positions. Finally, it executes a query to count the distinct positions, which returns the value 7.

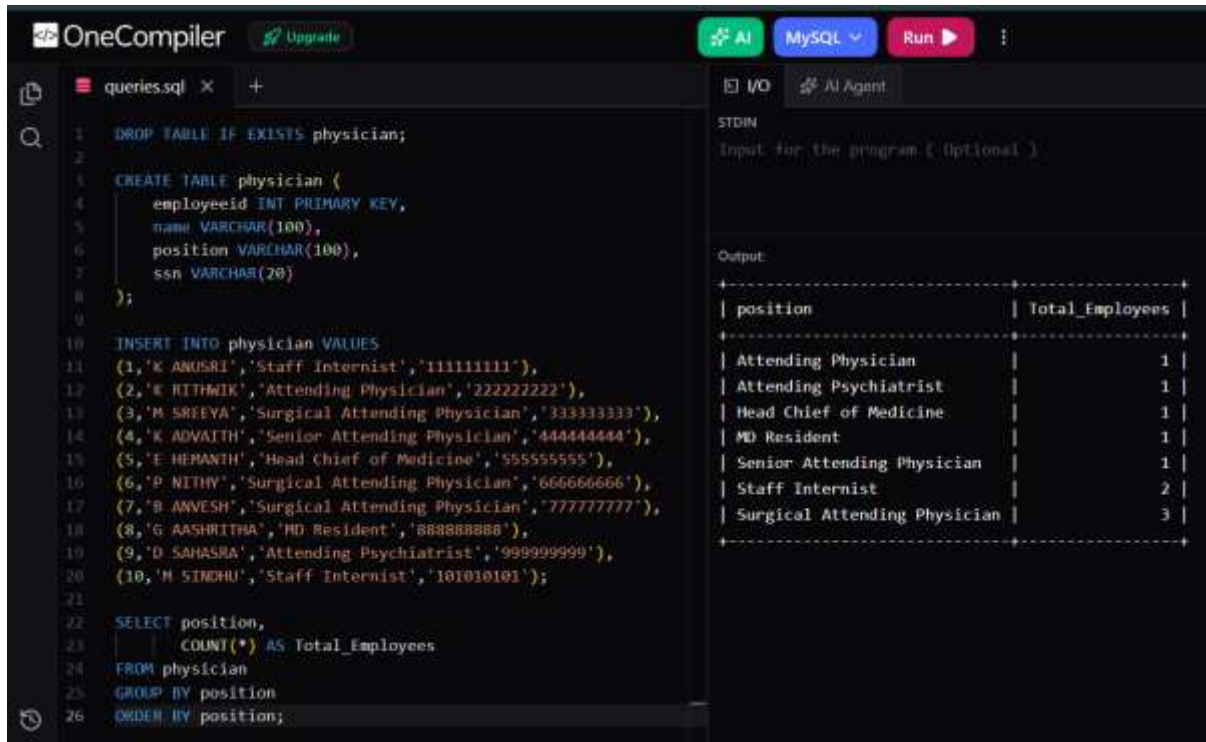
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','111111111'),
12 (2,'K RITHVIK','Attending Physician','222222222'),
13 (3,'M SREEYA','Surgical Attending Physician','333333333'),
14 (4,'K ADVAITH','Senior Attending Physician','444444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
16 (6,'P NITHY','Surgical Attending Physician','666666666'),
17 (7,'B ANVESH','Surgical Attending Physician','777777777'),
18 (8,'G AASHRITHA','MD Resident','888888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
20 (10,'M SINDHU','Staff Internist','101010101');
21
22 SELECT COUNT(DISTINCT position) AS Different_Positions
23 FROM physician;
```

Output:

Different_Positions
7

Q6 — GROUP BY

GROUP BY — Count Physicians in Each Position



The screenshot shows the OneCompiler MySQL interface. The left pane contains a SQL query file named 'queries.sql' with the following code:

```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHVIK','Attending Physician','22222222'),
13 (3,'M SREEVA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SINDHU','Staff Internist','10101010');
21
22 SELECT position,
23        COUNT(*) AS Total_Employees
24 FROM physician
25 GROUP BY position
26 ORDER BY position;
```

The right pane shows the output of the query, which is a table with two columns: 'position' and 'Total_Employees'.

position	Total_Employees
Attending Physician	1
Attending Psychiatrist	1
Head Chief of Medicine	1
MD Resident	1
Senior Attending Physician	1
Staff Internist	2
Surgical Attending Physician	3

Q7 — GROUP BY + HAVING

GROUP BY + HAVING — Find Positions with More Than One Physician

The screenshot shows the OneCompiler interface with a MySQL query and its output. The query creates a table named 'physician' with columns: employeeid (INT PRIMARY KEY), name (VARCHAR(100)), position (VARCHAR(100)), and ssn (VARCHAR(20)). It then inserts 10 rows of data. The final query is: `SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1;`

The output shows the following results:

position	Total_Employees
Staff Internist	2
Surgical Attending Physician	3

Q8 — ORDER BY

ORDER BY — Display Physicians in Alphabetical Order

The screenshot shows the OneCompiler interface with a MySQL query and its output. The query creates a table named 'physician' with columns: employeeid (INT PRIMARY KEY), name (VARCHAR(100)), position (VARCHAR(100)), and ssn (VARCHAR(20)). It then inserts 10 rows of data. The final query is: `SELECT * FROM physician ORDER BY name ASC;`

The output shows the following results (sorted alphabetically by name):

employeeid	name	position	ssn
7	B ANVESH	Surgical Attending Physician	77777777
9	D SAHASRA	Attending Psychiatrist	99999999
5	E HEMANTH	Head Chief of Medicine	55555555
8	G AASHRITHA	MD Resident	88888888
4	K ADVAITH	Senior Attending Physician	44444444
1	K ANUSRI	Staff Internist	11111111
2	K RITHMIK	Attending Physician	22222222
10	M SINDHU	Staff Internist	10101010
3	M SREEYA	Surgical Attending Physician	33333333
6	P NITHY	Surgical Attending Physician	66666666

Q9 — ORDER BY DESC

ORDER BY DESC — Display Physicians by Employee ID in Descending Order

The screenshot shows the OneCompiler interface with a MySQL database. The SQL code in the editor is as follows:

```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4   employeeid INT PRIMARY KEY,
5   name VARCHAR(100),
6   position VARCHAR(100),
7   ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','111111111'),
12 (2,'K RITHWIK','Attending Physician','222222222'),
13 (3,'M SREEYA','Surgical Attending Physician','333333333'),
14 (4,'K ADVAITH','Senior Attending Physician','444444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
16 (6,'P NITHY','Surgical Attending Physician','666666666'),
17 (7,'B ANVESH','Surgical Attending Physician','777777777'),
18 (8,'G AASHRITHA','MD Resident','888888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
20 (10,'M SINDHU','Staff Internist','101010101');
21
22 SELECT *
23 FROM physician
24 ORDER BY employeeid DESC;
```

The output on the right shows the results of the query, ordered by employeeid in descending order:

employeeid	name	position	ssn
10	M SINDHU	Staff Internist	101010101
9	D SAHASRA	Attending Psychiatrist	999999999
8	G AASHRITHA	MD Resident	888888888
7	B ANVESH	Surgical Attending Physician	777777777
6	P NITHY	Surgical Attending Physician	666666666
5	E HEMANTH	Head Chief of Medicine	555555555
4	K ADVAITH	Senior Attending Physician	444444444
3	M SREEYA	Surgical Attending Physician	333333333
2	K RITHWIK	Attending Physician	222222222
1	K ANUSRI	Staff Internist	111111111

Q10 — WHERE + IN

WHERE + IN — Find All Department Heads

The screenshot shows the OneCompiler interface with a MySQL database. The SQL code in the editor is as follows:

```
1 DROP TABLE IF EXISTS physician;
2 DROP TABLE IF EXISTS department;
3
4 CREATE TABLE physician (
5   employeeid INT PRIMARY KEY,
6   name VARCHAR(100),
7   position VARCHAR(100),
8   ssn VARCHAR(20)
9 );
10
11 CREATE TABLE department (
12   departmentid INT PRIMARY KEY,
13   name VARCHAR(100),
14   head INT
15 );
16
17 INSERT INTO physician VALUES
18 (1,'K ANUSRI','Staff Internist','111111111'),
19 (2,'K RITHWIK','Attending Physician','222222222'),
20 (3,'M SREEYA','Surgical Attending Physician','333333333'),
21 (4,'K ADVAITH','Senior Attending Physician','444444444'),
22 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
23 (6,'P NITHY','Surgical Attending Physician','666666666'),
24 (7,'B ANVESH','Surgical Attending Physician','777777777'),
25 (8,'G AASHRITHA','MD Resident','888888888'),
26 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
27 (10,'M SINDHU','Staff Internist','101010101');
28
29 INSERT INTO department VALUES
30 (1,'General Medicine',4),
31 (2,'Surgery',7),
32 (3,'Psychiatry',9);
33
34 SELECT *
35 FROM physician
36 WHERE employeeid IN (
37   SELECT head
38   FROM department
39 );
```

The output on the right shows the results of the query, which are the department heads:

employeeid	name	position	ssn
4	K ADVAITH	Senior Attending Physician	444444444
7	B ANVESH	Surgical Attending Physician	777777777
9	D SAHASRA	Attending Psychiatrist	999999999

Q11 — NOT IN / SUBQUERY

NOT IN — Find Physicians Who Are Not Department Heads

The screenshot shows the OneCompiler interface with a MySQL database. The SQL code in the editor defines two tables: `physician` and `department`. The `physician` table has columns `employeeid` (INT PRIMARY KEY), `name` (VARCHAR(100)), `position` (VARCHAR(100)), and `ssn` (VARCHAR(20)). The `department` table has columns `departmentid` (INT PRIMARY KEY), `name` (VARCHAR(100)), and `head` (INT). Data is inserted into both tables. The query uses a subquery to find department heads and then selects physicians whose `employeeid` is not in that list.

```
1 DROP TABLE IF EXISTS physician;
2 DROP TABLE IF EXISTS department;
3
4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10
11 CREATE TABLE department (
12     departmentid INT PRIMARY KEY,
13     name VARCHAR(100),
14     head INT
15 );
16
17 INSERT INTO physician VALUES
18 (1,'K AMURTI','Staff Internist','111111111'),
19 (2,'K RITHMEK','Attending Physician','222222222'),
20 (3,'M SREEYA','Surgical Attending Physician','333333333'),
21 (4,'K ADVAITH','Senior Attending Physician','444444444'),
22 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
23 (6,'P NITHY','Surgical Attending Physician','666666666'),
24 (7,'B ANVESH','Surgical Attending Physician','777777777'),
25 (8,'G AASHRITHA','MD Resident','888888888'),
26 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
27 (10,'M SINDHU','Staff Internist','101010101');
28
29 INSERT INTO department VALUES
30 (1,'General Medicine',4),
31 (2,'Surgery',7),
32 (3,'Psychiatry',9);
33
34 SELECT *
35 FROM physician
36 WHERE employeeid NOT IN (
37     SELECT head
38     FROM department
39 );
```

The output table shows the following data:

employeeid	name	position	ssn
1	K AMURTI	Staff Internist	111111111
2	K RITHMEK	Attending Physician	222222222
3	M SREEYA	Surgical Attending Physician	333333333
5	E HEMANTH	Head Chief of Medicine	555555555
6	P NITHY	Surgical Attending Physician	666666666
8	G AASHRITHA	MD Resident	888888888
10	M SINDHU	Staff Internist	101010101

Q12 — GROUP BY + ORDER BY

GROUP BY + ORDER BY — Rank Physician Positions by Employee Count

The screenshot shows the OneCompiler interface with a MySQL database. The SQL code defines the `physician` table and inserts data. The query uses `GROUP BY` to count the number of employees for each position and then `ORDER BY` to rank them in descending order of count.

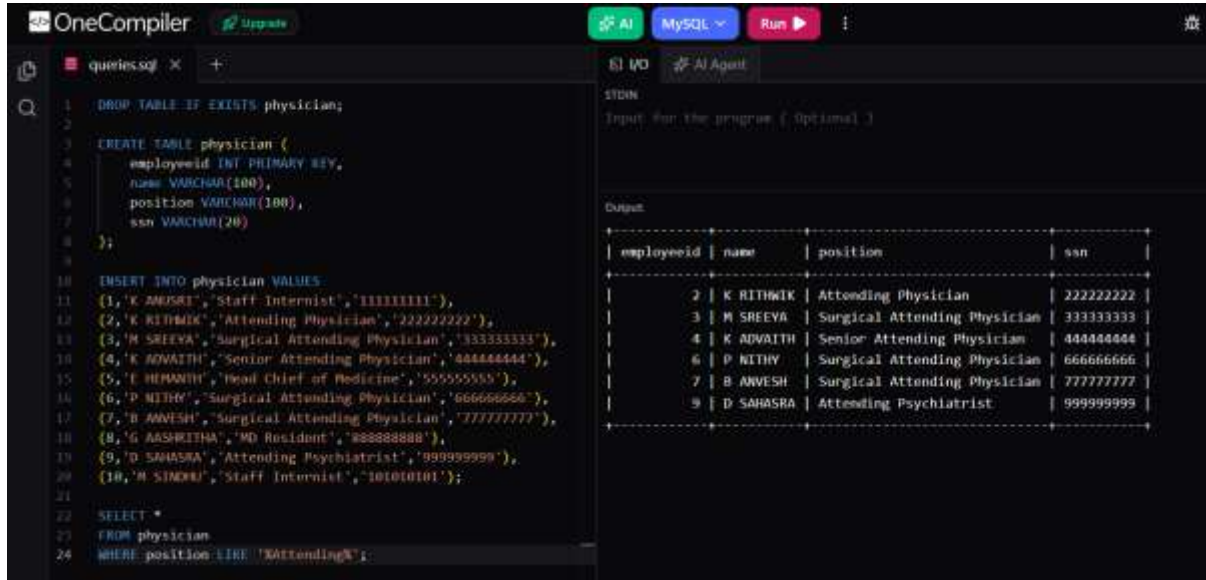
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K AMURTI','Staff Internist','111111111'),
12 (2,'K RITHMEK','Attending Physician','222222222'),
13 (3,'M SREEYA','Surgical Attending Physician','333333333'),
14 (4,'K ADVAITH','Senior Attending Physician','444444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
16 (6,'P NITHY','Surgical Attending Physician','666666666'),
17 (7,'B ANVESH','Surgical Attending Physician','777777777'),
18 (8,'G AASHRITHA','MD Resident','888888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
20 (10,'M SINDHU','Staff Internist','101010101');
21
22 SELECT position,
23        COUNT(*) AS Total_Employees
24 FROM physician
25 GROUP BY position
26 ORDER BY Total_Employees DESC;
```

The output table shows the following data:

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	2
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Q13 — WHERE + LIKE

LIKE — Find Physicians Whose Position Contains
“Attending”



The screenshot shows the OneCompiler MySQL IDE interface. The left pane contains a SQL script with the following code:

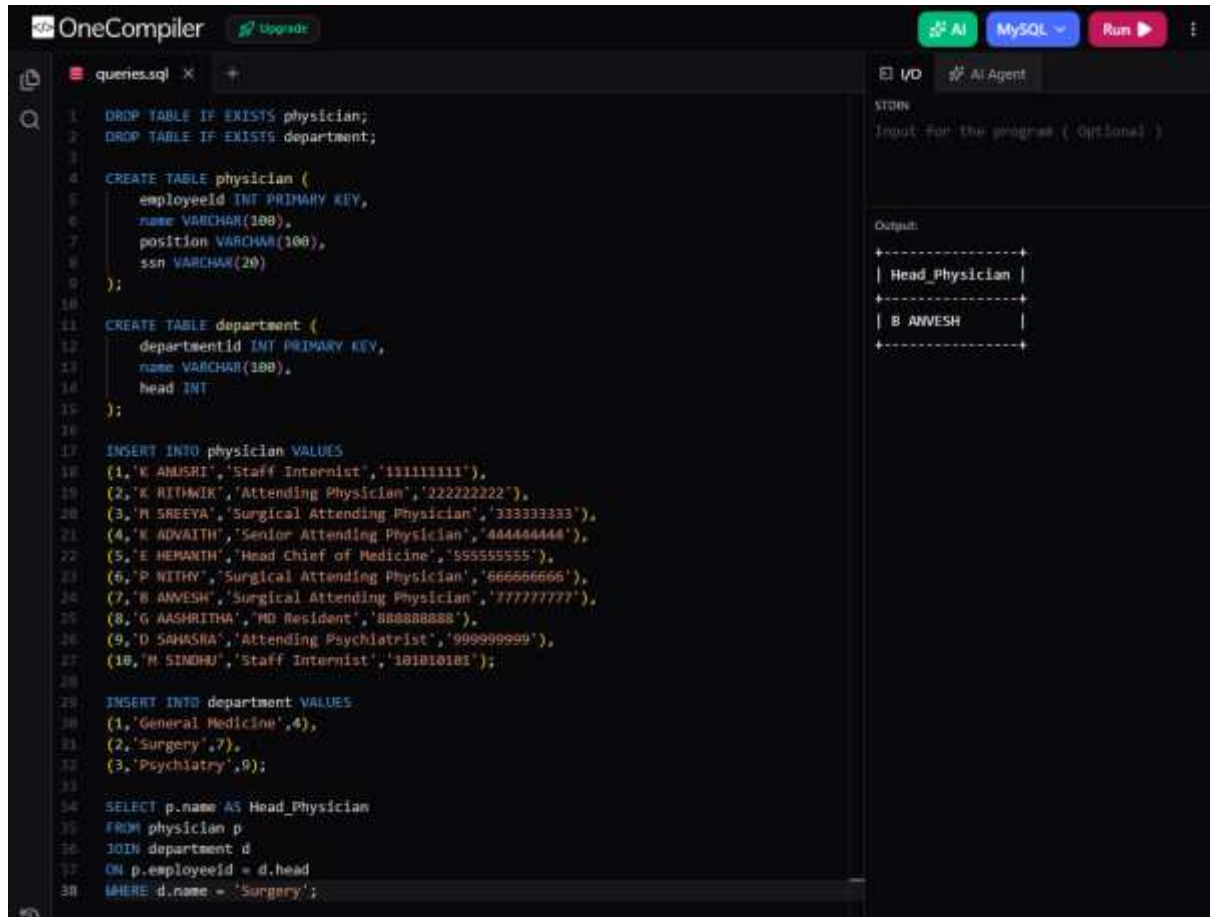
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeoid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHWIK','Attending Physician','22222222'),
13 (3,'M SREEYA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SIMRAN','Staff Internist','10101010');
21
22 SELECT *
23 FROM physician
24 WHERE position LIKE 'Attending%';
```

The right pane shows the output of the query, displaying a table with 4 columns: employeoid, name, position, and ssn. The results are as follows:

employeoid	name	position	ssn
2	K RITHWIK	Attending Physician	22222222
3	M SREEYA	Surgical Attending Physician	33333333
4	K ADVAITH	Senior Attending Physician	44444444
6	P NITHY	Surgical Attending Physician	66666666
7	B ANVESH	Surgical Attending Physician	77777777
9	D SAHASRA	Attending Psychiatrist	99999999

Q14 — JOIN + WHERE

JOIN + WHERE — Find the Head of the Surgery Department



The screenshot shows the OneCompiler IDE with a MySQL database. The code in the editor defines two tables: 'physician' and 'department'. The 'physician' table has columns for employeeid, name, position, and ssn. The 'department' table has columns for departmentid, name, and head. Data is inserted into both tables, and a query is executed to find the head of the Surgery department.

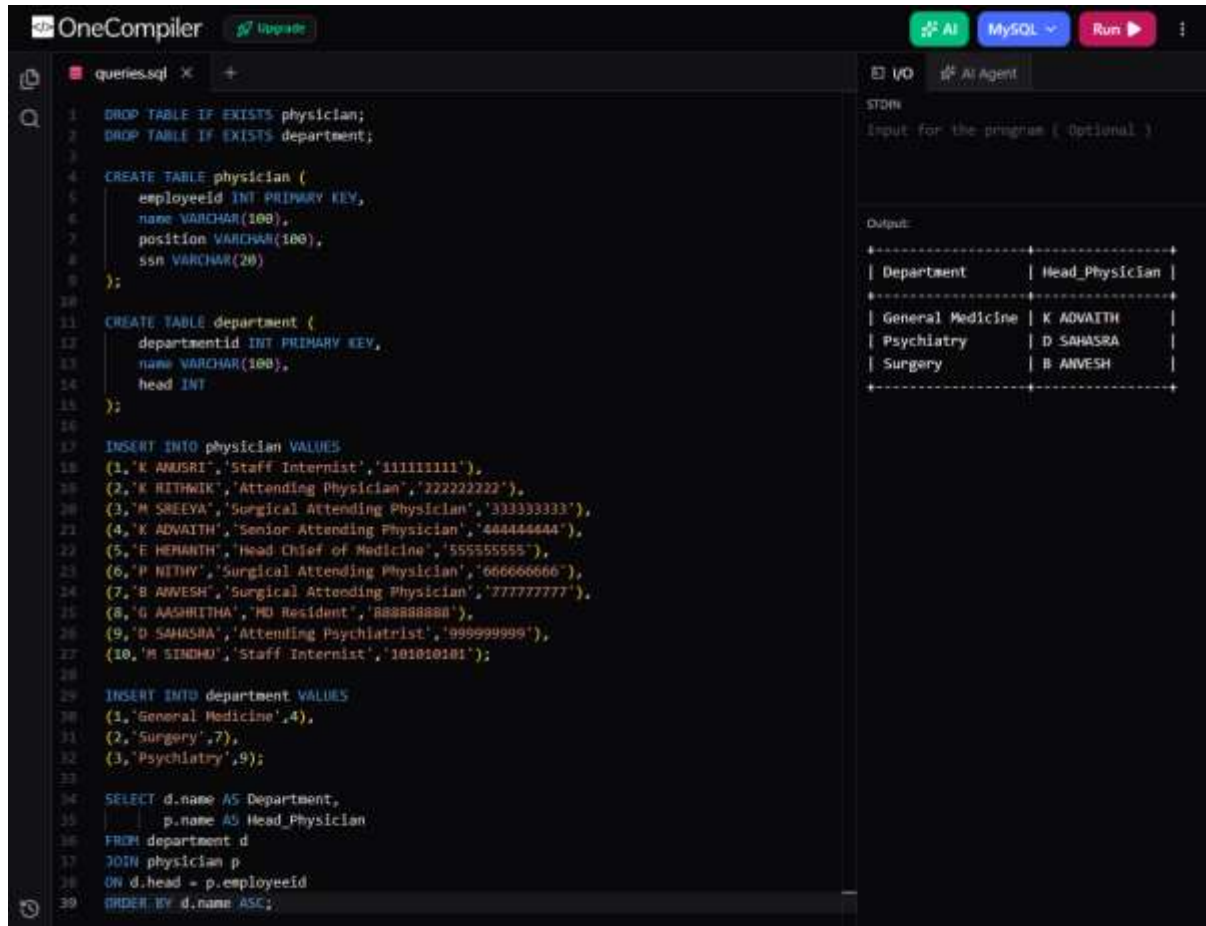
```
1 DROP TABLE IF EXISTS physician;
2 DROP TABLE IF EXISTS department;
3
4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10
11 CREATE TABLE department (
12     departmentid INT PRIMARY KEY,
13     name VARCHAR(100),
14     head INT
15 );
16
17 INSERT INTO physician VALUES
18 (1,'K. ANUSHI','Staff Internist','11111111'),
19 (2,'K. RITHVIK','Attending Physician','22222222'),
20 (3,'M. SREEVA','Surgical Attending Physician','33333333'),
21 (4,'K. ADVAITH','Senior Attending Physician','44444444'),
22 (5,'E. HEMANTH','Head Chief of Medicine','55555555'),
23 (6,'P. NITHY','Surgical Attending Physician','66666666'),
24 (7,'B. ANVESH','Surgical Attending Physician','77777777'),
25 (8,'G. AASHRITHA','MD Resident','88888888'),
26 (9,'D. SAHASRA','Attending Psychiatrist','99999999'),
27 (10,'M. SIMRAN','Staff Internist','10101010');
28
29 INSERT INTO department VALUES
30 (1,'General Medicine',4),
31 (2,'Surgery',7),
32 (3,'Psychiatry',9);
33
34 SELECT p.name AS Head_Physician
35 FROM physician p
36 JOIN department d
37 ON p.employeeid = d.head
38 WHERE d.name = 'Surgery';
```

Output:

Head_Physician
B ANVESH

Q15 — JOIN + ORDER BY

JOIN + ORDER BY — Display Departments and Their Heads in Alphabetical Order



The screenshot shows the OneCompiler MySQL IDE. The left pane contains a SQL script in `queries.sql` with line numbers 1 through 39. The script creates two tables, `physician` and `department`, inserts data into them, and then performs a join query to display department names and head physician names in alphabetical order.

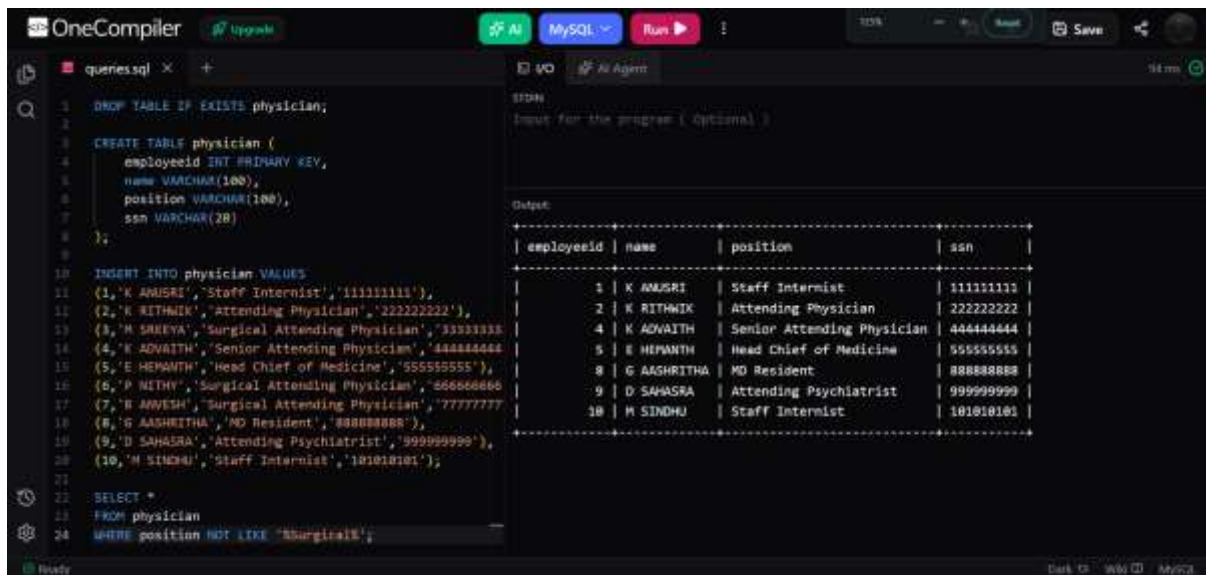
```
1 DROP TABLE IF EXISTS physician;
2 DROP TABLE IF EXISTS department;
3
4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10
11 CREATE TABLE department (
12     departmentid INT PRIMARY KEY,
13     name VARCHAR(100),
14     head INT
15 );
16
17 INSERT INTO physician VALUES
18 (1,'K AMUSRI','Staff Internist','111111111'),
19 (2,'K RITHVIK','Attending Physician','222222222'),
20 (3,'M SREEYA','Surgical Attending Physician','333333333'),
21 (4,'K ADVAITH','Senior Attending Physician','444444444'),
22 (5,'E HERANTH','Head Chief of Medicine','555555555'),
23 (6,'P NITHY','Surgical Attending Physician','666666666'),
24 (7,'B ANVESH','Surgical Attending Physician','777777777'),
25 (8,'G AASHRITHA','MD Resident','888888888'),
26 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
27 (10,'M SINDHU','Staff Internist','101010101');
28
29 INSERT INTO department VALUES
30 (1,'General Medicine',4),
31 (2,'Surgery',7),
32 (3,'Psychiatry',9);
33
34 SELECT d.name AS Department,
35        p.name AS Head_Physician
36 FROM department d
37 JOIN physician p
38 ON d.head = p.employeeid
39 ORDER BY d.name ASC;
```

The right pane shows the output of the query. It includes a status bar at the top with 'AI', 'MySQL', and 'Run' buttons. Below that, it says 'STDOUT' and 'Input for the program (Optional)'. The output is displayed in a table format:

Department	Head_Physician
General Medicine	K ADVAITH
Psychiatry	D SAHASRA
Surgery	B ANVESH

Q16 — NOT LIKE

NOT LIKE — Find Physicians Not Involved in Surgery



The screenshot shows the OneCompiler MySQL IDE. The left pane contains the following SQL code:

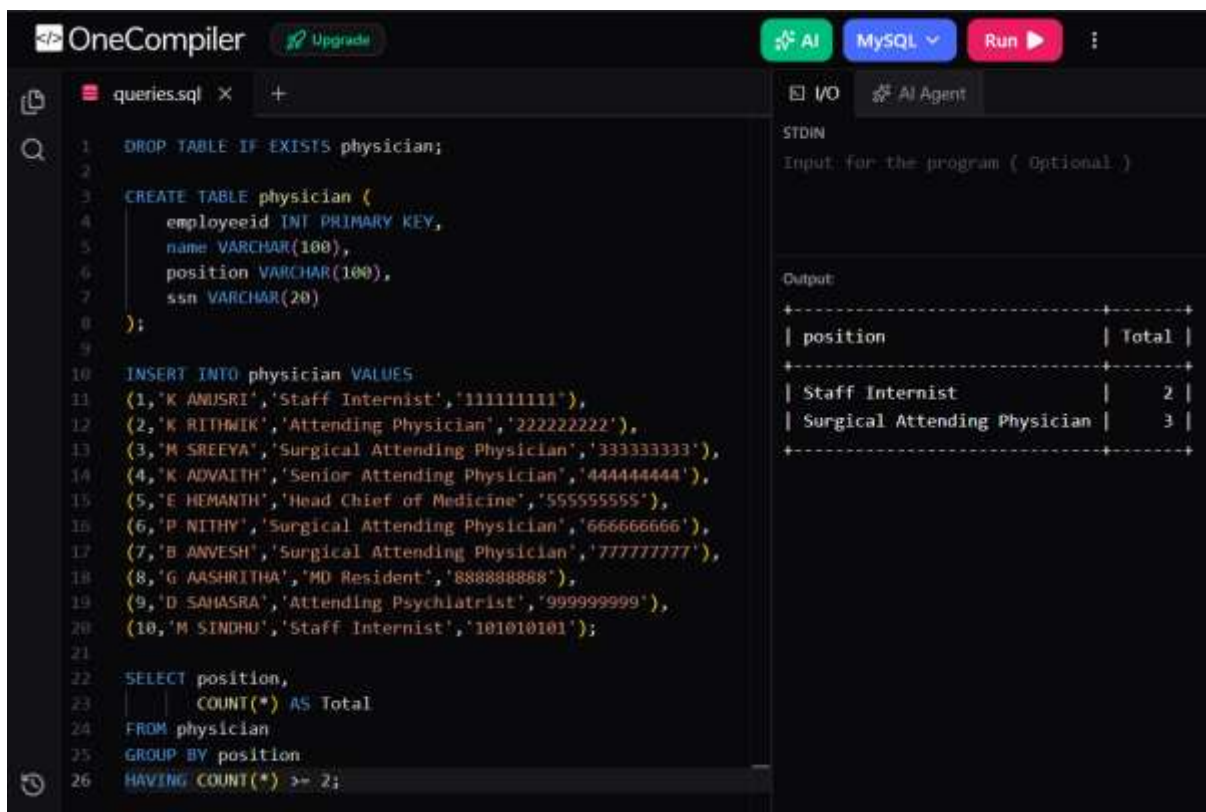
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHVIK','Attending Physician','22222222'),
13 (3,'M SREEYA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SINDHU','Staff Internist','10101010');
21
22 SELECT *
23 FROM physician
24 WHERE position NOT LIKE 'Surgical';
```

The right pane shows the output of the query:

employeeid	name	position	ssn
1	K ANUSRI	Staff Internist	11111111
2	K RITHVIK	Attending Physician	22222222
4	K ADVAITH	Senior Attending Physician	44444444
5	E HEMANTH	Head Chief of Medicine	55555555
8	G AASHRITHA	MD Resident	88888888
9	D SAHASRA	Attending Psychiatrist	99999999
10	M SINDHU	Staff Internist	10101010

Q17 — GROUP BY + HAVING

GROUP BY + HAVING — Find Positions Having at Least 2 Employees



The screenshot shows the OneCompiler MySQL IDE. The left pane contains the following SQL code:

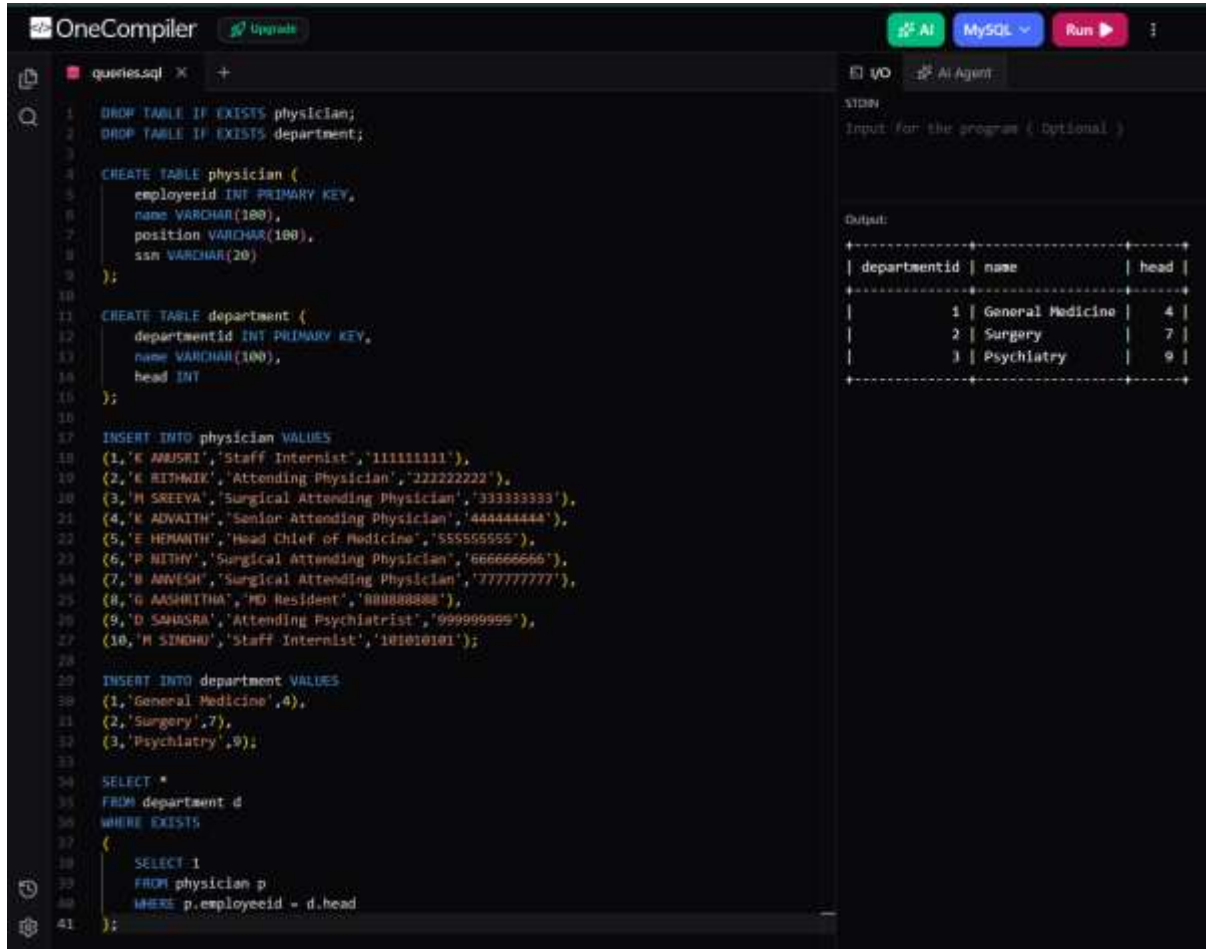
```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHVIK','Attending Physician','22222222'),
13 (3,'M SREEYA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SINDHU','Staff Internist','10101010');
21
22 SELECT position,
23        COUNT(*) AS Total
24 FROM physician
25 GROUP BY position
26 HAVING COUNT(*) >= 2;
```

The right pane shows the output of the query:

position	Total
Staff Internist	2
Surgical Attending Physician	3

Q18 — EXISTS

EXISTS — Find Departments That Currently Have a Valid Head



The screenshot shows the OneCompiler MySQL interface. The left pane contains a SQL script with the following queries:

```
1 DROP TABLE IF EXISTS physician;
2 DROP TABLE IF EXISTS department;
3
4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10
11 CREATE TABLE department (
12     departmentid INT PRIMARY KEY,
13     name VARCHAR(100),
14     head INT
15 );
16
17 INSERT INTO physician VALUES
18 (1,'K ANUSRI','Staff Internist','111111111'),
19 (2,'K RITHVIK','Attending Physician','222222222'),
20 (3,'M SREEVA','Surgical Attending Physician','333333333'),
21 (4,'K ADVAITH','Senior Attending Physician','444444444'),
22 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
23 (6,'P NITHY','Surgical Attending Physician','666666666'),
24 (7,'B ANVESH','Surgical Attending Physician','777777777'),
25 (8,'G AASHRITHA','MD Resident','888888888'),
26 (9,'D SAMSRA','Attending Psychiatrist','999999999'),
27 (10,'M SINDHU','Staff Internist','101010101');
28
29 INSERT INTO department VALUES
30 (1,'General Medicine',4),
31 (2,'Surgery',7),
32 (3,'Psychiatry',9);
33
34 SELECT *
35 FROM department d
36 WHERE EXISTS
37 (
38     SELECT 1
39     FROM physician p
40     WHERE p.employeeid = d.head
41 );
```

The right pane shows the output of the query:

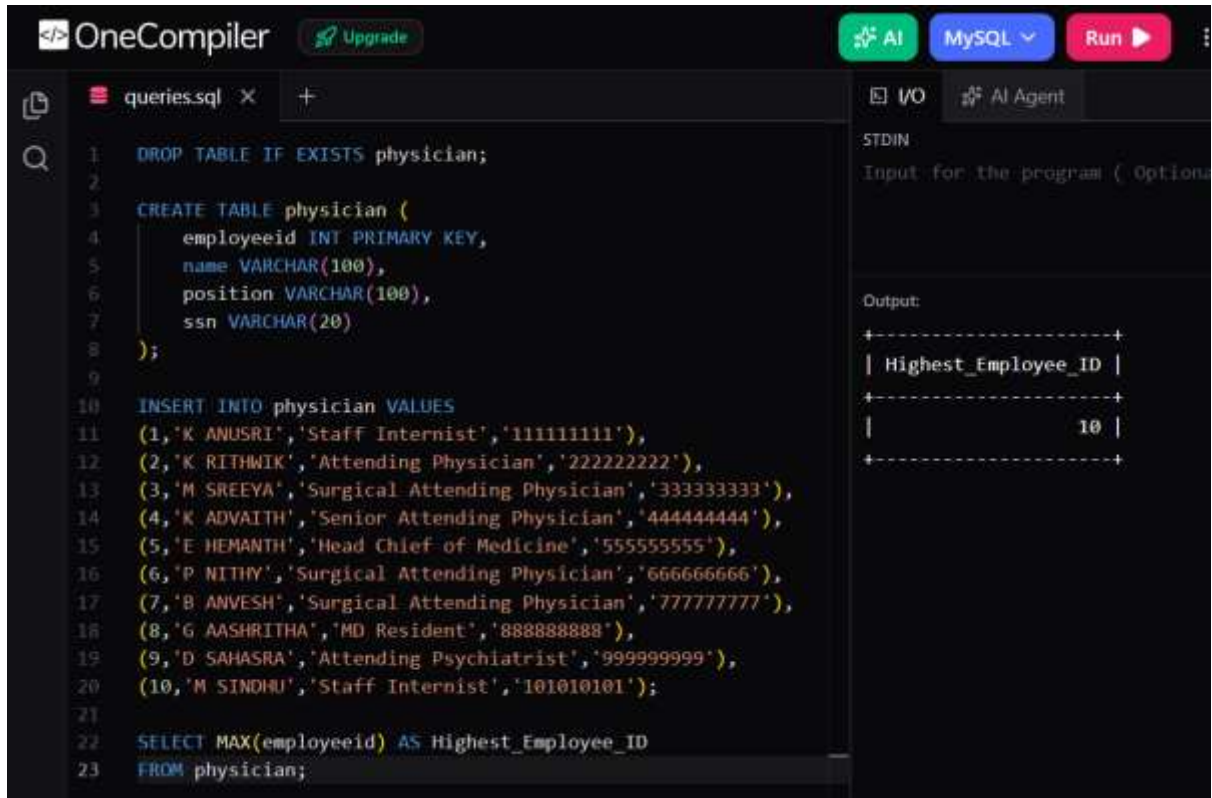
STDOUT
Input for the program (Optional)

Output:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

Q19 — MAX()

MAX() — Find the Highest Employee ID Currently Assigned



The screenshot shows the OneCompiler MySQL interface. The left pane contains a SQL query file named 'queries.sql' with the following code:

```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','11111111'),
12 (2,'K RITHMIK','Attending Physician','22222222'),
13 (3,'M SREEYA','Surgical Attending Physician','33333333'),
14 (4,'K ADVAITH','Senior Attending Physician','44444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','55555555'),
16 (6,'P NITHY','Surgical Attending Physician','66666666'),
17 (7,'B ANVESH','Surgical Attending Physician','77777777'),
18 (8,'G AASHRITHA','MD Resident','88888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','99999999'),
20 (10,'M SINDHU','Staff Internist','10101010');
21
22 SELECT MAX(employeeid) AS Highest_Employee_ID
23 FROM physician;
```

The right pane shows the output of the query:

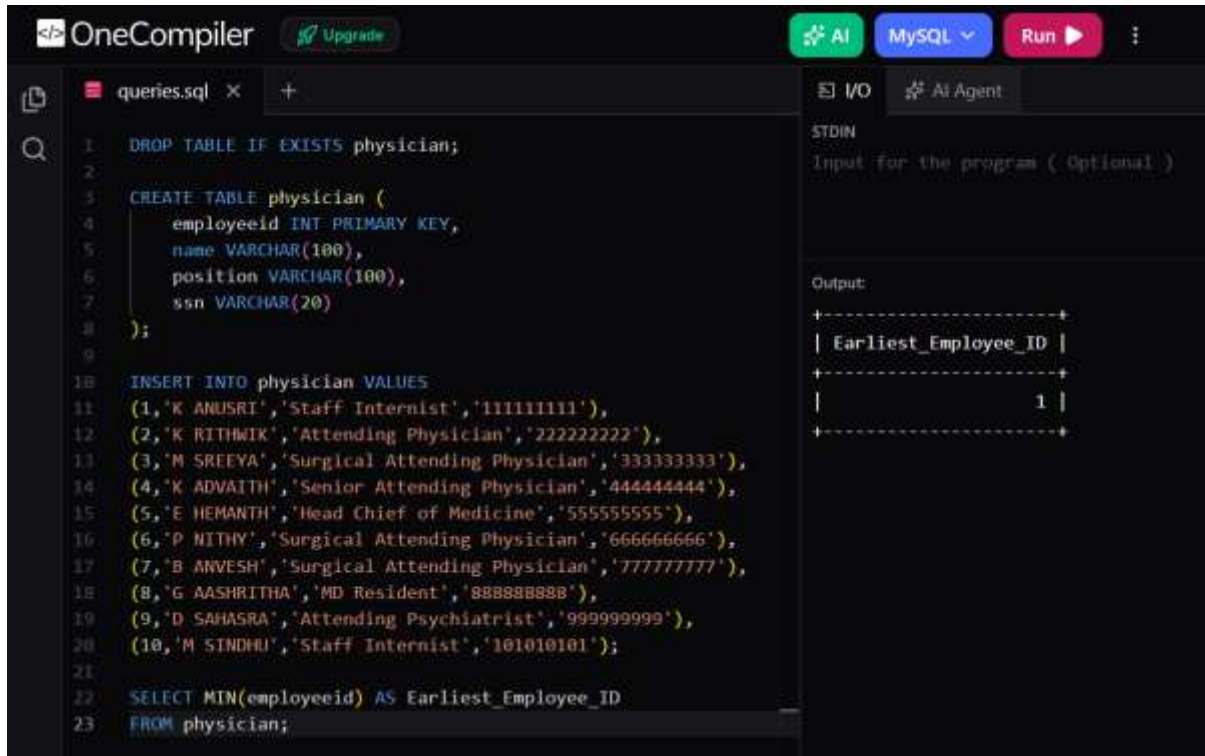
```
STDIN
Input for the program (Optional):

Output:
+-----+
| Highest_Employee_ID |
+-----+
|                    10 |
+-----+
```

\

Q20 — MIN()

MIN() — Find the Earliest Employee ID Currently Assigned



The screenshot shows the OneCompiler IDE interface. The left pane displays a SQL script in a file named 'queries.sql'. The script creates a table named 'physician' with columns: 'employeeid' (INT PRIMARY KEY), 'name' (VARCHAR(100)), 'position' (VARCHAR(100)), and 'ssn' (VARCHAR(20)). It then inserts 10 rows of data. The right pane shows the output of the query, which is a single row with the value '1' under the column header 'Earliest_Employee_ID'.

```
1 DROP TABLE IF EXISTS physician;
2
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K ANUSRI','Staff Internist','111111111'),
12 (2,'K RITHMIK','Attending Physician','222222222'),
13 (3,'M SREEYA','Surgical Attending Physician','333333333'),
14 (4,'K ADVAITH','Senior Attending Physician','444444444'),
15 (5,'E HEMANTH','Head Chief of Medicine','555555555'),
16 (6,'P NITHY','Surgical Attending Physician','666666666'),
17 (7,'B ANVESH','Surgical Attending Physician','777777777'),
18 (8,'G AASHRITHA','MD Resident','888888888'),
19 (9,'D SAHASRA','Attending Psychiatrist','999999999'),
20 (10,'M SINDHU','Staff Internist','101010101');
21
22 SELECT MIN(employeeid) AS Earliest_Employee_ID
23 FROM physician;
```

Output:

Earliest_Employee_ID
1